Dr. Shilpi Sengupta

Battery Cell Development Scientist | Materials & Electrochemical Engineering

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UK Right to Work: Global Talent visa – no employer sponsorship required

PROFESSIONAL PROFILE

Battery materials and cell development scientist with 2+ years of postdoctoral experience developing lithium-, sodium- and potassium-based energy-storage systems. Experienced in planning and executing systematic experimental programmes covering electrolyte and electrode materials, formulation, slurry processing, coating, drying, calendaring, cell assembly and electrochemical performance validation. Strong analytical background in EIS, CV, galvanostatic cycling, rate capability and degradation/performance analysis, with hands-on experience translating materials and process changes into cell-level performance improvements.

CORE BATTERY & ELECTROCHEMICAL CAPABILITIES

Cell & electrode development: Electrode slurry formulation and optimisation, planetary mixing, doctor-blade/automatic coating, drying and vacuum drying, calendaring/densification and 2032 coin-cell assembly.

Electrochemical testing & data interpretation: EIS, CV, LSV, galvanostatic cycling, rate capability, cycle-life testing, impedance analysis and equivalent-circuit modelling.

Electrolytes & interfaces: Inorganic solid electrolytes, ionic-liquid polymer electrolytes, polymer–ceramic composites, electrolyte films/pellets and electrode–electrolyte interface evaluation.

Materials characterisation: XRD, XPS, SEM, TEM, Raman, IR/FTIR, DSC/TGA and related structural/surface analysis to connect material properties with electrochemical performance.

Data & technical analysis: Origin, ZView, Bio-Logic/Origalys electrochemical analysis, TOPAS/XRD analysis, ImageJ and Python plotting.

Research delivery: Systematic experimental programmes, iterative/data-driven optimisation, technical documentation, multidisciplinary collaboration and mentoring of junior researchers.

Experimental programme planning & optimisation: systematic experimental design, comparative material evaluation, parameter optimisation and data-driven formulation development.

PROFESSIONAL EXPERIENCE

Postdoctoral Research Fellow | University of Birmingham, School of Chemistry | 2024–Present

• Design and optimise semi-solid, quasi-solid and all-solid-state electrolyte systems for lithium- and potassium-ion batteries, with emphasis on ionic transport, electrochemical stability and cell performance.

• Design and execute systematic electrochemical experiments to investigate ion-transport behaviour and electrode–electrolyte interfaces, using EIS, cyclic voltammetry, galvanostatic cycling and rate-performance measurements.

• Develop polymer–ceramic composite electrolytes and optimise formulations through iterative, data-driven experimentation to improve conductivity, cycling stability and interfacial compatibility.

• Fabricate and evaluate electrochemical cells under inert atmosphere, including electrolyte preparation, coin-cell assembly and electrochemical testing.
• Investigate performance-limiting mechanisms by correlating electrochemical behaviour with structural and surface analysis using XRD, XPS, SEM, TEM and Raman spectroscopy.

• Interpret electrochemical and materials datasets to identify formulation and processing changes, supporting evidence-based optimisation of electrolyte and cell performance.

• Work across multidisciplinary research activities, communicate findings through technical discussions and reporting, and supervise/mentor junior researchers.

Project Associate II | DST-MES Funded Project | SRMIST, CHENNAI, India |2019–2023

Project: Facile synthesis and development of 3D-graphene/WS_2/oxidized/CNT nanohybrids as advanced electrode materials for high-performance flexible supercapacitors.

• Developed 3D graphene/WS_2/CNT hybrid electrodes and optimised electrode architecture to improve energy density, cycling stability and flexibility.

• Designed ionic-liquid-based gel polymer electrolytes for high-voltage flexible devices and evaluated electrochemical performance.

• Fabricated flexible supercapacitor prototypes, taking materials and electrolyte formulations through device fabrication and validation.

• Contributed to a granted patent and multiple peer-reviewed publications.

SELECTED TECHNICAL PROJECT EVIDENCE

Solid-state & polymer electrolyte development — Developed inorganic and polymer–ceramic electrolyte systems for next-generation batteries; used electrochemical impedance and structural/surface analysis to understand ionic transport, interfaces and performance limitations.

Electrode processing & cell fabrication — Hands-on experience with slurry formulation, Thinky planetary mixing, doctor-blade and automatic coating, drying/vacuum drying, calendaring/densification and 2032 coin-cell assembly under controlled/inert conditions.

Flexible electrochemical device development — Took electrode and ionic-liquid polymer electrolyte formulations from material development through flexible prototype fabrication and electrochemical validation; contributed to patented technology.

PATENT & TECHNOLOGY OUTPUT

Granted Patent: High voltage and flexible supercapacitor with ionic liquid-based polymer gel electrolyte. Indian Patent Application No. 202241077229; publication 06.01.2023; Granted Patent No. 535352. Inventors: Shilpi Sengupta, Manab Kundu.

SELECTED PUBLICATIONS & RESEARCH OUTPUT

19 peer-reviewed research articles | 15 first/co-first author | Research across advanced electrode/electrolyte materials for lithium- and sodium-ion batteries and solid-state/flexible supercapacitors, including publications in Advanced Energy Materials, Small, Chemical Engineering Journal, ACS Applied Energy Materials, Energy Technology and Journal of Colloid and Interface Science.

Book Chapter: Sengupta, S.; Kundu, M. "Recent Advances of Sustainable Electrode Materials for Supercapacitor Devices", Sustainable Energy Storage in the Scope of Circular Economy, Wiley, 2023. ISBN 9781119817680.

EDUCATION

Ph.D. Chemistry — SRM Institute of Science and Technology, Kattankulathur, India | 2018–2023 Thesis: Nanostructured Transition Metal Derivatives for Electrochemical Energy Storage Applications.

M.Sc. Physical Chemistry — Savitribai Phule Pune University, Pune, India | 2013–2015 Grade A, 68.19%.

B.Sc. — Nagpur University, Nagpur, India | 2010–2013 Grade A, 67%.

SELECTED TECHNICAL TOOLS & EQUIPMENT

Gloveboxes (JACOMEX, MBraun), dry-room facilities, battery testers (Origalys OGF 500, Chentech, Bio-Logic VMP3), Solartron impedance analyser, 8-inch heat calendaring machine, compact tape-casting coater with vacuum chuck, film applicator/dryer; X'Pert HighScore Plus, TOPAS, ZView, Origin, ImageJ, XPS Peak, Vesta, CrystalMaker and Python.